

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Deep Learning, Generative AI Lab, LLM Applications, MLOps, Machine Learning	

SKILLS

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-learn
GenAI & LLMs: RAG, Prompt Engineering, Model Fine-Tuning, Semantic Search, Llama-3, GPT-4
MLOps & Vector DBs: MLflow, Milvus, FAISS, Neo4j, Docker, Prometheus, Grafana
Languages & Data: Python, SQL, NumPy, Pandas, C++, Java, Bash/Shell

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026

- **LLM Learning Engine:** Designed a learning engine using batch pattern extraction, GPT-4 L3 classification (92% precision), and confidence calibration to continuously improve AI recommendations at scale.
- **Hybrid Graph Search:** Built a 7-step pipeline combining Milvus vector similarity with Neo4j Cypher traversal, ranking remediation paths for ~20K exceptions/week with <200ms P95 latency.
- **Agentic Decision Capture:** Implemented closed-loop capture with DecisionTrace primitives, tracing agent decisions to graph relationships, improving recommendation accuracy by 15% over 8 weeks.
- **Evaluation & Drift Monitoring:** Created offline-online evaluation frameworks to track data drift and confidence scores, leveraging policy optimization to handle 3.5M records and \$14T in cash.

<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
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- **GenAI & Fine-tuning:** Engineered a micro-frontend translating natural language into JQL, fine-tuning Llama-3 (LoRA) on a 40K+ corpus to achieve a 94% Exact Match (EM) code generation rate.
- **RAG Pipeline:** Implemented Retrieval-Augmented Generation using Milvus vector stores and Cohere semantic reranking, boosting user acceptance to 82% and cutting query formulation time by 91%.

<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
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- **Predictive Modeling:** Built a scalable XGBoost workflow addressing manual configuration costs, achieving 88% precision and reducing manual intervention by 60%, deploying inferences with FastAPI.
- **Data Pipeline Engineering:** Orchestrated Apache Airflow pipelines to clean multi-terabyte transactional data with Pandas, uncovering latent patterns that optimized downstream routing efficiency.
- **MLOps Quality Engineering:** Built comprehensive Azure DevOps CI/CD pipelines with integrated JUnit testing suites, reducing deployment time-to-market by 40% and securing highly robust backend APIs.

Shunya	June 2020 – September 2020
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- Data Science Intern*
- **Computer Vision:** Engineered a GPU-accelerated PyTorch OpenCV workflow to analyze answer sheets, successfully automating evaluations and reducing manual grading time by 90%.
 - **Adaptive Learning:** Addressed student learning gaps by training K-means clustering models on multidimensional assessment data to dynamically recommend targeted practice questions.

PROJECTS

Production Recommendation & MLOps System

- Developed a hybrid recommender using LightFM and SVD++ (722 recs/s) with Llama-3-based cold-start logic, featuring a 4-hour retraining loop with zero-downtime reloading tracked via MLflow.
- **Model Auditing:** Built comprehensive model metric monitoring via Prometheus/Grafana and performed STRIDE, STPA, and demographic fairness audits prior to full production deployment.

Zulip Open-Source LLM Extensions

- Designed a cost-efficient Topic Drift Detection system backed by LLMs to correct conversation titles in real-time, optimizing interaction models to minimize API token usage costs.